

the capability to search and query, along with high throughput.

MySQL: This is an open source relational database management system that brings data consistency, scalability, high performance, availability and flexibility to IoT solutions by efficiently collecting data from IoT devices. It helps in data transformation through annotation and aggregation, enabling data to be understood in a better way. Based on the IoT solution's requirement, if the data format is fixed, then MySQL DB is the preferred choice.

GridDB: This is a container data model that extends the NoSQL key-value store. It represents data in the form of a collection referenced by keys. A transaction in GridDB guarantees ACID at the container level. It allows data to be present in two and three models. The two types of containers in GridDB are:

- *Collection container:* This is a general purpose container.
- *TimeSeries container:* This is data associated with a timestamp. It provides functions like data compression, data aggregation as well as high scalability, reliability and availability.

Redis: This is an in-memory open source database. It is a popular choice for IoT solutions as a hot database. It is widely used in IoT solutions for data ingest real-time analytics, messaging, caching and many other use cases. Today, it helps in edge computing involving deep learning, image recognition and other innovative computing requirements.

CrateDB: CrateDB is an open source distributed SQL database management system that fully integrates a searchable document-oriented data store. The CrateDB platform provides the distributed SQL query engine for faster joins, aggregations and ad-hoc queries. CrateDB is a highly scalable and available database. It supports various types of data.

Cassandra: This is a highly scalable and distributed open source database for managing voluminous

amounts of structured data across many commodity servers. It provides availability, linear scale performance, simplicity and easy distribution of data across multiple database servers. It supports strong data consistency across distributed architectures.

Hadoop: This is an open source software platform for distributed storage and distributed processing of very large data sets on computer clusters built from commodity hardware. It helps in driving analytics from all IoT data. It easily ingests data from multiple data sources and supports both batch as well as real-time data ingest from sensors using tools such as Apache Kafka and Apache Flume. It handles multiple IoT data types, structures and schemas, supporting real-time processing and applications on streaming data. Hadoop is a flexible, scalable and secure database. A number of automotive manufacturers, utilities, industrial automation companies, insurers, healthcare organisations, telecom and technology leaders are adopting Hadoop.

Benefits of open source databases in IoT

Open source databases have the following advantages over proprietary databases:

- Easily upgradable to new technologies.
- Have the capability to connect with upcoming device protocols and backend applications.
- Open source database adoption implies lower overall software cost, ease of change in technology and open source APIs for integration.
- Open source databases provide the flexibility to change the architecture of those solutions that are microservices centred.
- Flexibility to change the cloud service provider.

In order to make the right choice, we must first analyse the business problem and then arrive at the solution. We can then break up the solution into services, and understand the database needs for these services. This will help an organisation to narrow down the database choices.

Most IoT solutions can depend on a hot database for real-time data collection, processing, messaging and analytics. And cold databases can be depended on to store historical data and gather business intelligence. This will make the architecture simple, lean and robust. **END** 

References

- [1] <https://www.statista.com/statistics/471264/iot-number-of-connected-devices-worldwide/>
- [2] <https://www.gartner.com/en/newsroom/press-releases/2017-02-07-gartner-says-8-billion-connected-things-will-be-in-use-in-2017-up-31-percent-from-2016>
- [3] <https://www.ericsson.com/en/mobility-report/internet-of-things-forecast>

Acknowledgements

The authors would like to thank David Kenner and Raju Alluri of the Global Enterprise Architecture Group of Wipro Technologies for giving the required time and support to bring out this article as a part of the Architecture Practice efforts.

By: Dr Gopala Krishna Behara and Geeta Naik

Dr Gopala Krishna Behara is a distinguished member in the Global Enterprise Architecture division of Wipro. He has a total of 22 years of IT experience.
Geeta Naik is a senior enterprise architect in the Global Enterprise Architecture division of Wipro. She has 18 years of IT experience.

Disclaimer

The views expressed in this article are that of the authors and Wipro does not subscribe to their substance, veracity or truthfulness.